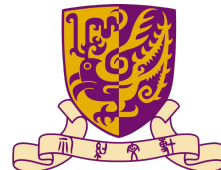
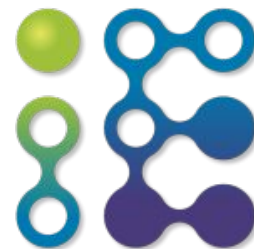


IERG4998, IERG4999

Suggested FYP topics for 2026-27

KSY1 ~ KSY6

Dr. Danny Yip
Department of Information Engineering



香港中文大學

The Chinese University of Hong Kong

MIEG students will not be accepted for any KSY projects in the first phase of matching.

If you are problem
students, you will have
a higher chance of
getting accepted. :)

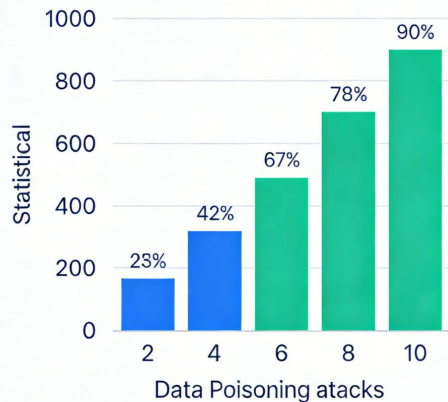
The following topics are reserved

- KSY1: AI-Powered Integrity Surveillance: Integrating Linguistic Obfuscation and Acoustic Prosody to Quantify Economic and Corporate Communications
- KSY2: Personalized Risk Stratification and Management Systems for the Future of Geriatric Care
- KSY3: A Multi-Model Ensemble System for Predicting Match Outcomes and Betting Value Identification
- KSY5: The Path to Superhuman Agency: A Comparative Analysis of Reinforcement Learning Agents in Gaming

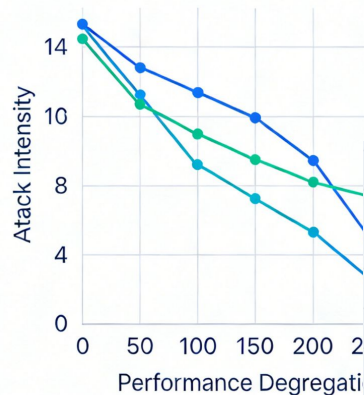
KSY4: Quantifying Resilience: A Comparative Analysis of Statistical and Data-Driven Approaches Against Data Poisoning Attacks

AI生成

Resilience Resilancs

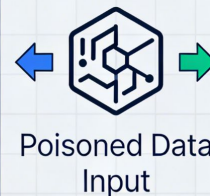
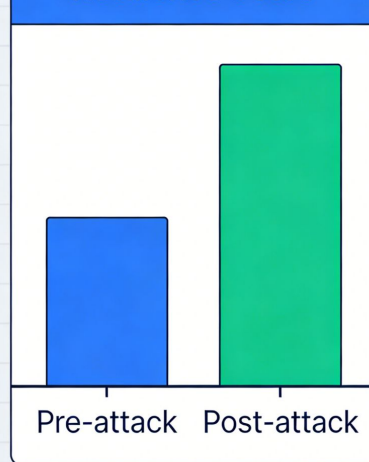


Data-Driven Approaches



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Resilience Score



Resilience Trend



Generated Figures

KSY4: Quantifying Resilience: A Comparative Analysis of Statistical and Data-Driven Approaches Against Data Poisoning Attacks

1. Systematize poisoning attack vectors.
2. Implement dual-stream defense frameworks.
3. Establish comparative resilience metrics.
4. Develop a resilience decision matrix.
5. Deploy defense mechanism onto real applications.
 - Quota for KSY4: 1 group project (2 students)

KSY6: Breakthroughs in Game Development

- A self-proposed group project relevant to game
- Pick your favourite game engine, e.g. Unity, Minecraft, Unreal, ... etc
- Dr. Yip will discuss some previous work of other groups with the students selecting this topic.
- Relevant job career: game developer
- Quota for KSY6: 1 group project (2 students)



Image credit: <https://unity.com/>, <https://www.minecraft.net/>,
<https://docs.unrealengine.com/4.27/en-US/Resources/>



KSY6: Breakthroughs in Game Development

- Students are expected to:
 1. Learn game programming.
 2. Master add-on, OOP, MVC, different design patterns.
 3. Learn state patterns on game control, e.g. lobby with multiple gaming maps.
 4. Set up a game server.
 5. Learn **reinforcement learning** for training game agents.
 6. Apply **recent technologies** for game development
 7. Record and edit videos on demonstrating your map.
 8. Identify **breakthroughs** on your game product.
- Warning: Writing game is not easy
- Warning: Writing game is not fun indeed

KSY6: Breakthroughs in Game Development (Examples)

Examples for okay directions:

1. Improved game agents in xxxxx
2. Improved game dynamics in xxxxx
3. Better game experience in xxxxx
4. Quick import of 3D game objects for game development
5. Apply xxxxx models in game

Examples for bad directions:

1. A Parkour map in xxxxx
2. Multiplayer Minesweeper